Opgaver Kapitel 6 og 7

Opgaver, Kapitel 6

6.50 44. Two people push a stalled car at its front doors, each applying a 280-N force at 25° to the forward direction, as shown in Fig. 6.16. How much work does each person do in pushing the car 5.5 m?

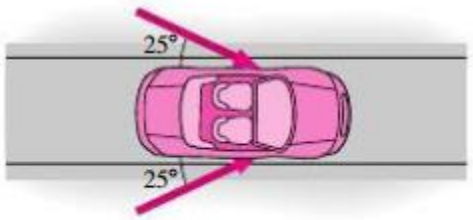

FIGURE 6.16 Problem 44

6.54 54. A force $\vec{F}$ acts in the x-direction, its magnitude given by $F = ax^2$, where x is in meters and $a = 3.0\ \mathrm{N/m^2}$. Find the work done by this force as it acts on a particle moving from $x = 0$ to $x = 6.0$ m.

6.73 73. You do 2.2 kJ of work pushing a 78-kg trunk at constant speed 3.1 m along a ramp inclined upward at 22°. What's the frictional coefficient between trunk and ramp?

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Opgaver, Kapitel 7

7.37

33. A more accurate expression for the force law of the rope in Example 7.3 is $F = -kx + bx^2 - cx^3$, where k and b have the values given in Example 7.3 and $c = 3.1\ \text{N/m}^3$. Find the energy stored in stretching the rope 2.62 m. By what percentage does your result differ from that of Example 7.3?

7.45

43. A 100-g block slides back and forth on a frictionless surface between two springs, as shown in Fig. 7.18. The left-hand spring has $k = 110$ N/m and its maximum compression is 20 cm. The right-hand spring has $k = 240$ N/m. Find (a) the maximum compression of the right-hand spring and (b) the speed of the block as it moves between the springs.

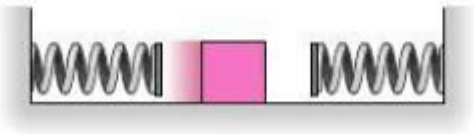

FIGURE 7.18 Problem 43

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